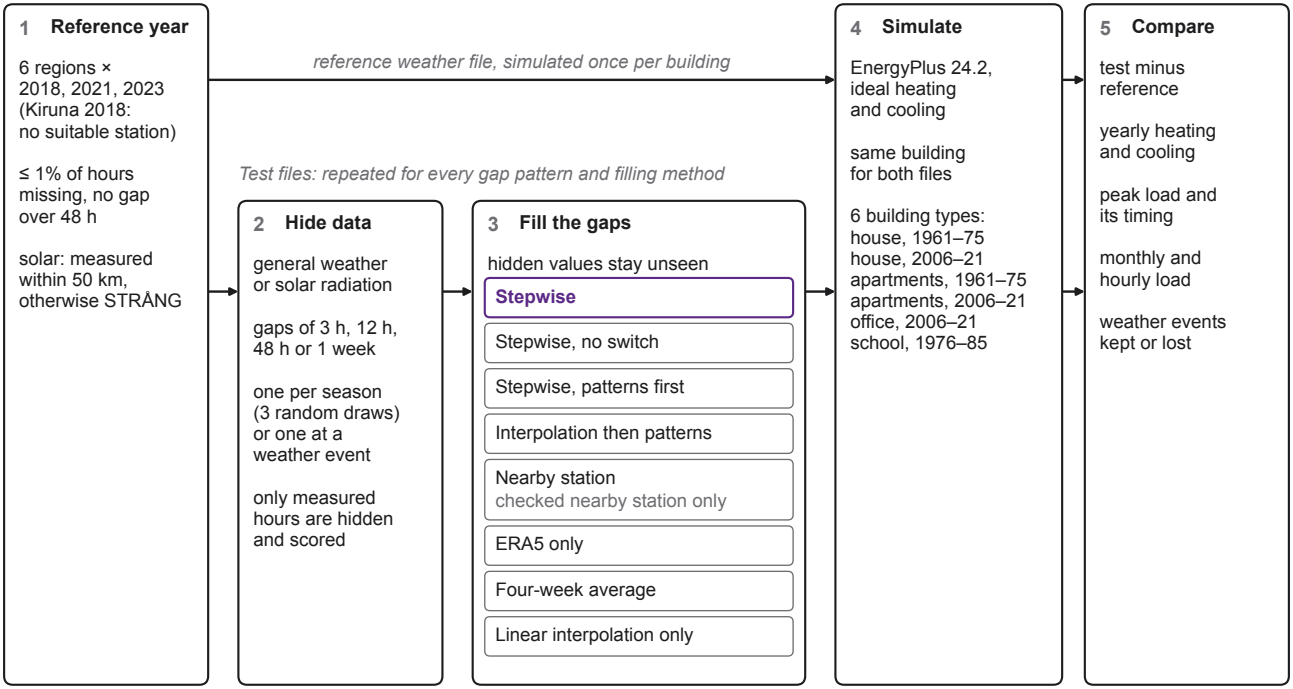

1 Reference year
6 regions × 2018, 2021, 2023 (Kiruna 2018: no suitable station)
≤ 1% of hours missing, no gap over 48 h
solar: measured within 50 km, otherwise STRÅNG
reference weather file, simulated once per building
Test files: repeated for every gap pattern and filling method
2 Hide data
general weather or solar radiation
gaps of 3 h, 12 h, 48 h or 1 week
one per season (3 random draws) or one at a weather event
only measured hours are hidden and scored
3 Fill the gaps
hidden values stay unseen
Stepwise
Stepwise, no switch
Stepwise, patterns first
Interpolation then patterns
Nearby station
checked nearby station only
ERA5 only
Four-week average
Linear interpolation only
4 Simulate
EnergyPlus 24.2, ideal heating and cooling
same building for both files
6 building types:
house, 1961–75
house, 2006–21
apartments, 1961–75
apartments, 2006–21
office, 2006–21
school, 1976–85
5 Compare
test minus reference
yearly heating and cooling
peak load and its timing
monthly and hourly load
weather events kept or lost